

Artificial General Intelligence and the Sectoral Transition to a Post-Labor Economy: A Dynamic General Equilibrium Analysis

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Abstract

This paper develops a dynamic general equilibrium model to analyze the macroeconomic and distributive consequences of artificial general intelligence (AGI) diffusion. Building on task-based and sectorally heterogeneous production, we model AGI capital as a general-purpose technology that substitutes asymmetrically for human labor across sectors and skill classes. The model incorporates endogenous AGI investment, nonlinear diffusion dynamics, and labor displacement, along with a fiscal mechanism that taxes AGI-generated value added and redistributes it via targeted universal basic income (UBI). We show that AGI adoption leads to a secular decline in labor income and a rising concentration of capital rents, generating structural inequality. Absent intervention, the economy converges to a post-labor steady state marked by vanishing wages and increasing income concentration. However, an AGI-linked value-added tax, rebated to displaced workers through targeted UBI, can fully offset income losses and improve intertemporal welfare. Despite transitional frictions due to capital adjustment costs, the model admits a balanced growth path with sustained output growth in the absence of human labor. These results provide a tractable framework for evaluating automation-era fiscal architecture and offer policy-relevant insights into redistribution, labor displacement, and structural macroeconomic adjustment.

keywords: Artificial General Intelligence (AGI); Automation; Universal Basic Income (UBI); Value-Added Tax (VAT); General Equilibrium; Labor Displacement

1 Introduction

The accelerating progress in artificial general intelligence (AGI) promises to redefine the structure of production, the nature of labor, and the foundations of economic distribution. Unlike task-specific automation or narrow AI, AGI entails the development of broadly capable, adaptable systems that can substitute for human labor across a wide spectrum of cognitive and manual tasks (Russell and Norvig, 2010; Goertzel, 2007; Brynjolfsson and

McAfee, 2014; Korinek, 2022). The implications of this shift for employment, growth, and inequality are profound and demand new frameworks that go beyond conventional modeling of labor-displacing technologies.

A large body of work has analyzed the economic impact of automation. Pioneering task-based models by Acemoglu and Restrepo (2018, 2020, 2022) demonstrate that automation can reduce labor demand and shift the labor income share while boosting productivity. Empirical studies such as Autor (2015), Graetz and Michaels (2018), and Bloom et al. (2020) show that automation contributes to labor market polarization and rising inequality, with significant sectoral heterogeneity. However, these studies typically treat automation as exogenous or partial, and do not address the economy-wide implications of general-purpose AGI technologies.

In contrast to earlier work emphasizing task-level automation, recent macroeconomic research has begun to engage directly with the broader implications of artificial general intelligence (AGI). Korinek (2021), Korinek and Stiglitz (2021), and Stiefenhofer (2025) explore how AGI may induce large-scale technological unemployment, raising profound distributional and institutional challenges. Trajtenberg (2018) characterizes AI as a general-purpose technology (GPT) with wide-ranging spillover effects across sectors, while Brynjolfsson et al. (2022) emphasize its potential to raise productivity through complementary innovations. Models of endogenous automation, such as Guerreiro et al. (2020), formalize firms' technology adoption decisions as equilibrium responses to relative prices and factor substitution. These approaches illuminate the general equilibrium impact of automation but typically abstract from redistributive policy design and sectoral heterogeneity. Stiefenhofer and Chen (2024) extend this literature by analyzing the dynamic stability of industrial artificial intelligence within Cobb–Douglas production structures, offering early insight into the long-run macroeconomic behavior of AGI-driven economies.

Several new contributions further highlight the urgency of analyzing AGI's systemic impact. OECD (2024) and IMF (2024) provide institutional assessments of AI's macroeconomic and distributional effects. Acemoglu (2024) offers a stylized macroeconomic framework capturing key substitution and complementarity forces driven by AGI. Windmann et al. (2024) and Lee and Su (2025) explore how industrial AI challenges integration into existing systems, while Erdil et al. (2025) propose a novel dynamic model (GATE) linking AI, growth, and redistribution. Subaveerapandiyana and Shimray (2024) analyze the bibliometric evolution of AI-labor studies, emphasizing the field's expanding scope.

In the context of general equilibrium modeling, automation has been integrated into heterogeneous agent frameworks such as Krusell and Smith (2000), who analyze inequality dynamics, and Hubmer et al. (2020), who study the role of capital taxation and wealth concentration. Jones (2016) and Guner et al. (2023) extend endogenous growth models to examine the macroeconomic effects of AI and automation. A related strand of work focuses on the design of redistributive institutions under automation, including proposals for universal basic income (UBI) (Bidadanure, 2019; Hoynes and Rothstein, 2019; Gentilini, 2020). Werner (2021) and Lowrey (2018) explore how UBI interacts with technological displacement and labor market participation. However, there is limited work linking AGI-induced productivity gains to financing mechanisms for redistribution. Notable exceptions include Koenig et al. (2022), who incorporate optimal transfers in a GE setting, and Korinek (2021), who model the fiscal implications of taxing AI-generated surplus.

To advance this literature, we adopt a dynamic, sectorally heterogeneous general equilibrium framework in which AGI capital is both a productive and taxable input. A general equilibrium model is especially suited to studying AGI because it captures how labor displacement, capital reallocation, and policy interventions—such as value-added taxation and universal basic income—interact across markets and sectors simultaneously (Daruich and Fernández, 2021; Imrohoroglu et al., 2021; Nir et al., 2021). Unlike partial equilibrium approaches, GE analysis allows for endogenous feedbacks in factor prices, income distribution, and fiscal flows, which are essential when technological change is pervasive and nonlinear (Jones, 2021; Lofgren et al., 2002). For example, Daruich and Fernández (2021) analyze a UBI-financed transition in a dynamic GE model with automation shocks, showing that transition dynamics critically affect intergenerational welfare. Similarly, Imrohoroglu et al. (2021) and Nir et al. (2021) use GE frameworks to explore how automation and policy affect sectoral labor displacement, capital accumulation, and occupational choice. These studies demonstrate that a general equilibrium setting is indispensable for capturing both the short-run disruption and long-run restructuring induced by AGI. In particular, the GE structure accommodates sector-specific automation dynamics, general-purpose technology spillovers, and the reallocation of displaced labor and capital—making it a robust platform for analyzing both transitional and steady-state outcomes under AGI diffusion.

AGI diffusion is modeled via sector-specific logistic functions, generating time-varying exposure to automation across labor classes. Output is produced through a nested CES aggregator combining human labor, AGI labor (modeled as capital-embedded services), and two forms of capital. We introduce a value-added tax (VAT) on AGI-attributed output and use the proceeds to finance a targeted universal basic income calibrated to displaced labor. Importantly, we incorporate nonlinear AGI capital adjustment costs and derive transitional dynamics consistent with empirical observations of technological diffusion (Comin and Hobijn, 2010; Bresnahan and Trajtenberg, 1995).

Our contribution builds on task-based frameworks (Acemoglu and Restrepo, 2019; Restrepo, 2022), general equilibrium analyses of inequality (Krusell and Smith, 2000; Hubmer et al., 2020), and macro-fiscal redistribution models (Korinek, 2021; Aksoy et al., 2023). In contrast to prior work that treats AGI as a static productivity shift or exogenous technology, we explicitly embed AGI in the structure of production, finance, and income distribution. We formally derive conditions under which AGI-VAT revenues can sustain income replacement for displaced labor, characterize the emergence of balanced growth without human labor, and highlight transitional disequilibria induced by capital adjustment frictions.

By embedding AGI into both the production and fiscal architecture of a dynamic economy, the model provides a platform for analyzing long-run welfare outcomes, inequality trajectories, and the fiscal viability of universal transfers. It also raises normative questions about the role of the state in governing AGI diffusion and mediating its gains, particularly in economies where income becomes increasingly decoupled from labor (Piketty, 2014; Farhi and Werning, 2019).

This paper is organized as follows. Section 2 introduces the dynamic general equilibrium model featuring AGI diffusion, value-added taxation (VAT), and targeted universal basic income (UBI). Section 3 presents the analytical results, focusing on automation dynamics, distributional shifts, and transitional general equilibrium properties. Section 4 discusses the policy implications and situates the findings within the existing academic literature. Section

5 concludes. Technical results and some proofs are provided in the Appendix.

2 Model Overview

We consider a continuous-time economy over $t \in [0, \infty)$, populated by agents partitioned into labor classes $j \in \{c, m, h\}$ and economic sectors $i = 1, \dots, n$. The total population is given by

$$N = \sum_{j \in \{c, m, h\}} N_j, \quad (1)$$

where N_j denotes the size of labor class j : cognitive workers ($j = c$, early automation risk), manual workers ($j = m$, late automation risk), and hybrid workers ($j = h$, intermediate risk). Labor is distributed across sectors according to the matrix $\beta = [\beta_{ij}]$, where $\beta_{ij} \geq 0$ is the fraction of labor class j employed in sector i , satisfying

$$\sum_i \beta_{ij} = 1, \quad \forall j. \quad (2)$$

Automation intensity in sector i follows a logistic diffusion process

$$\alpha_i(t) = \frac{1}{1 + e^{-\kappa_i(t-t_i^*)}}, \quad (3)$$

where $\kappa_i > 0$ is the sector-specific diffusion rate and t_i^* is the inflection point. This functional form is supported by empirical studies on the adoption of general-purpose technologies such as electricity, IT, and AI (Griliches, 1957; Mansfield, 1961; Comin and Hobijn, 2010). It captures slow initial adoption, rapid acceleration, and eventual saturation. The automation exposure of labor class j is a weighted average of sectoral automation intensities

$$\gamma_j(t) = \sum_i \beta_{ij} \alpha_i(t). \quad (4)$$

The effective labor supply of class j in sector i is then

$$L_{ij}(t) = \beta_{ij} N_j (1 - \gamma_j(t)), \quad (5)$$

reflecting that $\gamma_j(t)$ reduces the productive share of labor due to automation. Total effective human labor in sector i is

$$L_{i,h}(t) = \sum_j L_{ij}(t). \quad (6)$$

AGI labor input in sector i is

$$L_{i,AGI}(t) = \psi_i \alpha_i(t) K_i^{(AGI)}(t), \quad (7)$$

where $\psi_i > 0$ is a productivity parameter, $K_i^{(AGI)}(t)$ is AGI-specific capital, and $\alpha_i(t)$ determines its activation level. This expression defines AGI labor not as a traditional labor input, but as a derived, capital-embedded service flow that substitutes for human labor.

Specifically, $K_i^{(AGI)}(t)$ represents the stock of AGI-enabled capital (e.g., autonomous systems, intelligent software), and $\alpha_i(t)$ denotes the sector's automation intensity—governing the degree to which this capital is operationalized. The parameter $\psi_i > 0$ captures task-specific or sectoral productivity differences in translating AGI capital into effective labor services. The structure reflects the idea that AGI labor is activated capital: without automation infrastructure ($\alpha_i(t)$), AGI capital is idle; but as automation spreads, AGI capital increasingly delivers labor-equivalent functionality. This approach aligns with task-based models of automation (Acemoglu and Restrepo, 2019) and extends capital–labor substitution by introducing endogenous, technology-driven activation of digital labor inputs. Unlike traditional labor, AGI labor is not constrained by population, but by the pace of technological diffusion and investment in specialized capital.

Production in sector i is modeled using a three-level nested CES aggregator:

$$Q_i(t) = A_i \left[\theta_1 (L_{i,h}(t)^{\rho_1} + L_{i,AGI}(t)^{\rho_1})^{\frac{\eta_1}{\rho_1}} + \theta_2 \left((K_i^{(H)}(t))^{\rho_2} + (K_i^{(AGI)}(t))^{\rho_2} \right)^{\frac{\eta_2}{\rho_2}} \right]^{\frac{1}{\eta}}, \quad (8)$$

where $A_i > 0$ denotes total factor productivity, and $\theta_1 + \theta_2 = 1$ are share parameters. The parameters ρ_1 and ρ_2 govern the curvature of the respective CES nests for labor and capital inputs. The inner substitution elasticities are given by $1/(1 - \eta_1)$ and $1/(1 - \eta_2)$, capturing the substitutability between human and AGI labor and between traditional and AGI-specific capital, respectively. The outer elasticity parameter η regulates the substitutability between the two composite bundles—labor and capital—allowing for sectoral flexibility in the allocation of value added between factor types. This three-layered structure captures both the within-factor substitution effects and the broader reallocation dynamics that characterize AGI-driven automation, consistent with empirical CES specifications in the growth and technology diffusion literature.

The nested CES structure reflects a nuanced representation of production under AGI-driven automation. The innermost labor nest captures the idea that human labor $L_{i,h}(t)$ and AGI labor $L_{i,AGI}(t)$ are imperfect substitutes in performing tasks. The substitution parameter ρ_1 governs the elasticity between these inputs, enabling the model to span a continuum between automation-augmenting environments—where human and AGI labor are only weakly substitutable—and automation-replacing regimes, in which AGI increasingly displaces human input (Acemoglu and Restrepo, 2019).

Parallel to this, the capital nest aggregates traditional physical capital $K_i^{(H)}(t)$ and AGI-specific capital $K_i^{(AGI)}(t)$. This formulation acknowledges the growing role of intelligent capital assets, such as autonomous systems and machine learning platforms, in production. The elasticity parameter ρ_2 determines the degree of substitution between conventional and AGI-embedded capital, aligning with frameworks in which capital heterogeneity responds endogenously to technological progress (Acemoglu and Restrepo, 2022).

At the outer level, the model combines the labor and capital composites through a CES aggregator governed by the elasticity parameter η . This outer nest allows for variation in how sectors substitute between total labor-type and capital-type inputs, accommodating structural asymmetries and sectoral diversity in the adoption of automation. The form is

consistent with two-level CES specifications commonly used in empirical studies of production and technological change (Krusell and Smith, 2000).

The share parameters θ_1 and θ_2 , which satisfy $\theta_1 + \theta_2 = 1$, determine the relative weight of the labor and capital nests in sectoral value added. These weights capture differences in production composition across sectors, such as between labor-intensive services and capital-intensive manufacturing, and facilitate comparative statics as automation and capital accumulation evolve over time.

This production architecture is particularly well-suited for modeling AGI as a labor-equivalent input that is nonetheless capital-embodied. By explicitly nesting AGI labor alongside human labor and pairing AGI-specific capital with traditional capital, the model captures the dual character of intelligent systems: as both cognitive substitutes for human labor and as physical assets that require capital investment. This design integrates the model with task-based theories of automation and endogenous technological change, while remaining tractable within calibrated macroeconomic frameworks.

In each sector i , output prices are determined under the standard assumption of perfect competition. Firms behave as price takers and set prices equal to marginal cost. Let $C_i(Q_i(t))$ denote the total cost function associated with producing quantity $Q_i(t)$, which accounts for expenditures on effective labor, traditional capital, AGI-specific capital, and any sector-specific adjustment costs. The marginal cost is then defined as the derivative of the total cost with respect to output:

$$MC_i(t) = \frac{\partial C_i(Q_i(t))}{\partial Q_i(t)}. \quad (9)$$

Under the zero-profit condition characteristic of competitive equilibrium, firms set prices equal to marginal cost:

$$P_i(t) = MC_i(t) = \frac{\partial C_i(Q_i(t))}{\partial Q_i(t)}. \quad (10)$$

This pricing rule follows directly from microeconomic duality theory. Specifically, it can be derived from the firm's cost minimization problem via Shepard's Lemma

$$\frac{\partial C_i(Q_i(t), \mathbf{p}(t))}{\partial Q_i(t)} = \lambda(t), \quad (11)$$

where $\lambda(t)$ is the Lagrange multiplier on the production constraint and represents the marginal cost of output. The input price vector $\mathbf{p}(t)$ includes the cost of both traditional and AGI-related inputs, making this formulation sensitive to changes in automation and technological intensity. In the nested CES framework introduced in Equation (8), the marginal cost function is not available in closed form due to the nonlinear interactions among labor types, capital types, and their substitution elasticities. Instead, $MC_i(t)$ is implicitly defined and varies with the sectoral automation intensity, the degree of factor substitutability, and the evolving composition of capital. Accordingly, numerical methods—such as local log-linearization or calibrated simulations—are typically employed to characterize price dynamics and analyze equilibrium responses to policy or technological shocks.

Capital accumulation in each sector i is governed by the following differential equations

$$\dot{K}_i^{(H)}(t) = I_i^{(H)}(t) - \delta_i^{(H)} K_i^{(H)}(t), \quad (12)$$

$$\dot{K}_i^{(AGI)}(t) = I_i^{(AGI)}(t) - \phi_i \left(I_i^{(AGI)}(t), K_i^{(AGI)}(t) \right), \quad (13)$$

where $K_i^{(H)}(t)$ and $K_i^{(AGI)}(t)$ denote the sector-specific stocks of traditional and AGI-related capital, respectively, and $I_i^{(H)}(t)$, $I_i^{(AGI)}(t)$ are the corresponding gross investments. The parameter $\delta_i^{(H)} \in (0, 1)$ is the constant depreciation rate for traditional capital. In contrast, the evolution of AGI capital reflects nonlinear adjustment dynamics. Depreciation in this case is modeled as a convex function of both investment and capital

$$\phi_i(I, K) = \frac{\vartheta_i}{2} \left(\frac{I}{K} \right)^2 K = \frac{\vartheta_i}{2} \frac{I^2}{K}, \quad (14)$$

where $\vartheta_i > 0$ is a sector-specific adjustment cost parameter. This functional form implies that the effective loss of AGI capital increases disproportionately with the rate of investment relative to the existing capital stock. To see this formally, note that for fixed $K > 0$, the depreciation function $\phi_i(I, K)$ is strictly convex in I , as the second derivative satisfies

$$\frac{\partial^2 \phi_i(I, K)}{\partial I^2} = \frac{\vartheta_i}{K} > 0.$$

This convexity captures the frictions inherent in AGI deployment—such as organizational restructuring, compatibility constraints, training requirements, and coordination delays—which make rapid scaling of AGI capital increasingly costly. These dynamics are consistent with the adjustment cost literature on general-purpose technologies (Hayashi, 1982; Acemoglu and Restrepo, 2022). The use of a nonlinear depreciation function implies that investment decisions must account not only for marginal capital returns but also for endogenous capital losses due to scaling inefficiencies. This enriches the model's dynamics and allows it to capture realistic transitional frictions in economies undergoing structural automation.

Households in labor class j derive utility from consumption and face income fluctuations due to sector-specific automation. Lifetime utility is given by the discounted logarithmic functional form

$$U_j = \int_0^\infty e^{-\rho t} \log(C_j(t)) dt, \quad (15)$$

where $C_j(t)$ denotes consumption at time t , and $\rho > 0$ is the constant rate of time preference. Households maximize utility subject to a period-by-period budget constraint

$$C_j(t) \leq (1 - \gamma_j(t))w_j(t) + \text{UBI}(t), \quad (16)$$

where $\gamma_j(t) \in [0, 1]$ represents the automation exposure of labor class j , $w_j(t)$ is the class-specific wage rate, and $\text{UBI}(t)$ is a universal basic income transfer financed through taxation on automated production. The term $(1 - \gamma_j(t))$ reflects the employed share of the population within class j , with higher $\gamma_j(t)$ indicating greater technological displacement. Aggregated consumption across class j is given by

$$C_j^{\text{agg}}(t) = N_j [(1 - \gamma_j(t))w_j(t) + \text{UBI}(t)], \quad (17)$$

where N_j is the total mass of individuals in labor class j . For tractability, we assume that all physical and AGI-specific capital is owned by firms. Households do not earn capital income; their total income derives exclusively from wages and UBI transfers. This assumption isolates the distributional effects of automation to the labor market channel and the redistribution mechanism, abstracting from heterogeneity in asset holdings. Wages are determined by the marginal productivity of human labor

$$w_j(t) = \sum_i \beta_{ij} \frac{\partial Q_i(t)}{\partial L_{i,h}(t)}, \quad (18)$$

where β_{ij} denotes the exposure weight of labor class j to sector i , and $Q_i(t)$ is the production function in sector i . This specification implies that a worker's earnings depend on their sectoral allocation and the sector-specific marginal product of human labor. To finance redistribution, we introduce a value-added tax (VAT) on AGI-enabled production. The VAT is designed to target automation-derived surplus, ensuring that firms deploying AGI contribute in proportion to its productive role. The VAT rate in sector i is specified as

$$\tau_i(t) = \tau_0 \cdot \frac{C_{i,AGI}(t)}{P_i(t)Q_i(t)}, \quad (19)$$

where $\tau_0 \in [0, 1]$ is a policy parameter, $C_{i,AGI}(t)$ denotes the cost of AGI inputs in sector i , and $P_i(t)Q_i(t)$ is the total value of sectoral output. This structure ensures that VAT burdens scale with the degree of automation, targeting sectors where AGI capital contributes a larger share of value-added. Total VAT revenue collected from sector i is

$$T_i(t) = \tau_i(t)P_i(t)Q_i(t), \quad T(t) = \sum_i T_i(t), \quad (20)$$

where $T(t)$ is the aggregate VAT revenue available for redistribution. We assume that these funds are transferred exclusively to displaced workers. The total mass of displaced individuals is

$$N_{\text{disp}}(t) = \sum_j N_j \gamma_j(t), \quad (21)$$

and the UBI transfer per displaced individual is given by

$$\text{UBI}(t) = \begin{cases} \frac{T(t)}{N_{\text{disp}}(t)}, & \text{if } N_{\text{disp}}(t) > 0, \\ 0, & \text{otherwise.} \end{cases} \quad (22)$$

To guarantee the adequacy of this policy, we impose a fiscal constraint requiring that total VAT revenue must at least replace the wage income lost due to displacement

$$T(t) \geq \sum_j N_j \gamma_j(t) w_j(t), \quad (23)$$

which implies that

$$\text{UBI}(t) \geq \frac{1}{N_{\text{disp}}(t)} \sum_j N_j \gamma_j(t) w_j(t).$$

This ensures that each displaced worker receives a transfer approximating their forgone wage income, maintaining baseline consumption levels and sustaining aggregate demand during the automation transition. The mechanism aligns with policy proposals advocating for automation-adjusted fiscal instruments to manage distributional effects and preserve macroeconomic stability (Acemoglu and Restrepo, 2022; Korinek, 2021; Guner et al., 2023).

Definition 2.1 (Dynamic General Equilibrium). *A dynamic general equilibrium is a collection of time-dependent functions*

$$\{C_j(t), P_i(t), w_j(t), Q_i(t), K_i^{(H)}(t), K_i^{(AGI)}(t), I_i^{(H)}(t), I_i^{(AGI)}(t), T_i(t), \text{UBI}(t)\}_{i,j,t \geq 0}$$

such that for all $t \geq 0$, the following conditions hold:

(i) **Household optimization:** For each labor class j ,

$$\{C_j(t)\} = \arg \max_{C_j(\cdot)} \int_0^\infty e^{-\rho t} \log C_j(t) dt$$

subject to the present-value intertemporal budget constraint

$$\int_0^\infty e^{-\rho t} C_j(t) dt \leq \int_0^\infty e^{-\rho t} [(1 - \gamma_j(t))w_j(t) + \text{UBI}(t)] dt.$$

(ii) **Firm profit maximization:** For each sector i , firms choose capital inputs $\{K_i^{(H)}(t), K_i^{(AGI)}(t)\}$ and investment $\{I_i^{(AGI)}(t)\}$ to maximize

$$\begin{aligned} \max_{\{K_i^{(H)}(t), K_i^{(AGI)}(t)\}} \int_0^\infty e^{-rt} \left\{ P_i(t) Q_i(t) - \sum_j \beta_{ij} N_j (1 - \gamma_j(t)) w_j(t) \right. \\ \left. - r_H(t) K_i^{(H)}(t) - r_{AGI}(t) K_i^{(AGI)}(t) - \phi_i \left(I_i^{(AGI)}(t), K_i^{(AGI)}(t) \right) \right\} dt \end{aligned} \quad (24)$$

subject to the sectoral production function

$$Q_i(t) = A_i \left[\theta_1 (L_{i,h}(t)^{\rho_1} + L_{i,AGI}(t)^{\rho_1})^{\frac{\eta_1}{\rho_1}} + \theta_2 \left((K_i^{(H)}(t))^{\rho_2} + (K_i^{(AGI)}(t))^{\rho_2} \right)^{\frac{\eta_2}{\rho_2}} \right]^{\frac{1}{\eta}}.$$

(iii) **Capital accumulation:**

$$\dot{K}_i^{(H)}(t) = I_i^{(H)}(t) - \delta_i^{(H)} K_i^{(H)}(t), \quad \dot{K}_i^{(AGI)}(t) = I_i^{(AGI)}(t) - \phi_i \left(I_i^{(AGI)}(t), K_i^{(AGI)}(t) \right).$$

(iv) **Labor consistency:**

$$L_{ij}(t) = \beta_{ij} N_j (1 - \gamma_j(t)), \quad L_{i,h}(t) = \sum_j L_{ij}(t), \quad L_{i,AGI}(t) = \psi_i \alpha_i(t) K_i^{(AGI)}(t).$$

(v) **Wage determination:**

$$w_j(t) = \sum_i \beta_{ij} \frac{\partial Q_i(t)}{\partial L_{i,h}(t)}.$$

(vi) **Goods market clearing:**

$$\sum_i P_i(t) Q_i(t) = \sum_j N_j C_j(t) + \sum_i \left[I_i^{(H)}(t) + I_i^{(AGI)}(t) \right].$$

(vii) **Government budget balance:**

$$T(t) = \text{UBI}(t) \cdot N_{\text{disp}}(t), \quad N_{\text{disp}}(t) = \sum_j N_j \gamma_j(t).$$

(viii) **Price setting:**

$$P_i(t) = \frac{\partial C_i(Q_i(t), \mathbf{p}(t))}{\partial Q_i(t)}.$$

This definition formalizes a dynamic general equilibrium framework designed to capture the structural transformation of an economy undergoing automation and the integration of artificial general intelligence (AGI). Households maximize intertemporal utility under income constraints that reflect declining labor earnings due to technological displacement, partially offset by redistributive transfers in the form of universal basic income (UBI). The distribution of income across labor classes is shaped by heterogeneous exposure to automation, introducing dynamic inequality that the fiscal mechanism is designed to mitigate.

Firms make intertemporal investment decisions over traditional and AGI-specific capital stocks, subject to convex adjustment costs that reflect real-world frictions in deploying general-purpose technologies. These firms operate in perfectly competitive markets, setting output prices equal to marginal cost. The production structure incorporates a nested CES technology that allows for imperfect substitution between human and AGI labor, and between conventional and AGI-specific capital, thereby capturing the evolving factor intensity of production as automation deepens.

The government imposes a value-added tax (VAT) on the AGI-attributed share of sectoral output. Revenues from this tax are recycled into a targeted UBI scheme that compensates technologically displaced workers. This mechanism operationalizes a core insight from the literature on automation and fiscal externalities: labor-displacing technologies reduce wage-based tax bases while concentrating value-added in capital-intensive production, thereby justifying automation-contingent instruments (Acemoglu and Restrepo, 2022; Broer et al., 2022; Korinek, 2021). By linking VAT rates to AGI-related cost shares, the model ensures that more automated sectors contribute proportionally more to redistribution—without directly taxing capital investment—thus preserving incentives for innovation while internalizing the distributional costs of automation (Guner et al., 2023; Aksoy et al., 2023).

A fiscal adequacy constraint requires that VAT revenues are sufficient to cover the average forgone labor income of displaced workers. This condition ensures consumption smoothing during the transition to a post-labor economy, aligning with normative arguments in the

macro-distributional literature for automation-adjusted transfer schemes (Farhi and Werning, 2020; Acemoglu and Restrepo, 2019). Notably, AGI labor is modeled as a capital-embedded service flow rather than a wage-earning agent, reinforcing the disconnect between productivity gains and human income in advanced automation regimes and highlighting the redistributive role of policy.

Overall, the model contributes to the literature on general-purpose technologies (GPTs), endogenous technological diffusion, and fiscal policy under structural change (Acemoglu and Restrepo, 2018; Comin and Hobijn, 2010; Krusell and Smith, 2000). It provides a theoretically grounded, policy-relevant framework for evaluating the interaction between automation, taxation, and redistribution under realistic conditions of sectoral heterogeneity and labor-market asymmetry. By explicitly incorporating an automation-linked VAT and a displacement-calibrated UBI, the model offers a coherent and tractable basis for designing equitable transitions in economies confronting AGI-induced disruption.

3 Model Analysis

This section develops key theoretical results that characterize the economic dynamics and policy implications arising from the diffusion of artificial general intelligence (AGI) across sectors and labor classes. We analyze how heterogeneous exposure to automation creates asymmetries in labor displacement and discuss the fiscal mechanisms—such as value-added taxation on AGI output—available to capture the economic surplus generated by automation. The analysis further explores the sustainability of redistribution policies like universal basic income (UBI) funded by AGI-related tax revenues, highlighting their potential to stabilize welfare and inequality in a post-labor economy. Through a series of propositions and rigorous proofs, we establish fundamental insights into the growth trajectories, transitional consumption dynamics, and welfare outcomes characterizing the transition to an economy dominated by AGI capital and labor. These results provide a formal foundation for designing targeted, class-sensitive policy interventions to manage the disruptive impacts of automation and ensure equitable economic outcomes.

Proposition 3.1 (Asymmetry in Automation Exposure Across Labor Classes). *Let $\alpha_i(t) = \frac{1}{1+e^{-\kappa_i(t-t_i^*)}}$ denote AGI intensity in sector i , and define the automation exposure of labor class j as*

$$\gamma_j(t) = \sum_{i=1}^n \beta_{ij} \alpha_i(t), \quad (25)$$

where $\beta_{ij} \geq 0$ and $\sum_{i=1}^n \beta_{ij} = 1$ for all j . If two labor classes $j_1 \neq j_2$ differ in sectoral exposure vectors $\beta_{j_1} \neq \beta_{j_2}$, then there exists $t > 0$ such that $\gamma_{j_1}(t) \neq \gamma_{j_2}(t)$. That is, automation exposure is strictly asymmetric across labor classes with different sectoral weights.

Proof. We restate the definition of AGI intensity in sector i at time t

$$\alpha_i(t) = \frac{1}{1 + e^{-\kappa_i(t-t_i^*)}}, \quad (26)$$

a logistic function with parameters $\kappa_i > 0$ and inflection point $t_i^* \in \mathbb{R}$. This function satisfies

$$\alpha_i(t) \in (0, 1) \quad \text{for all } t \in \mathbb{R}, \quad (27)$$

$$\lim_{t \rightarrow -\infty} \alpha_i(t) = 0, \quad \lim_{t \rightarrow \infty} \alpha_i(t) = 1. \quad (28)$$

Each $\alpha_i(t)$ is strictly increasing, continuous, and differentiable in t . Automation exposure for labor class j is defined as a convex combination of sectoral intensities

$$\gamma_j(t) = \sum_{i=1}^n \beta_{ij} \alpha_i(t), \quad (29)$$

where $\beta_{ij} \geq 0$ and $\sum_{i=1}^n \beta_{ij} = 1$ for all j . Let $j_1 \neq j_2$ be two labor classes with distinct exposure vectors: $\beta_{j_1} \neq \beta_{j_2}$. Then there exists at least one sector i^* such that

$$\delta_i := \beta_{ij_1} - \beta_{ij_2} \neq 0 \quad \text{for some } i = i^*. \quad (30)$$

Define the difference in automation exposure between the two labor classes

$$\Delta(t) := \gamma_{j_1}(t) - \gamma_{j_2}(t) = \sum_{i=1}^n \delta_i \alpha_i(t), \quad (31)$$

where $\delta_i = \beta_{ij_1} - \beta_{ij_2}$. Since both weight vectors sum to one, we have $\sum_i \delta_i = 0$, implying that $\Delta(t)$ is a linear combination of $\alpha_i(t)$ with zero-sum weights. Because at least one $\delta_i \neq 0$ and the $\alpha_i(t)$ are strictly increasing, nonlinear, and parametrically heterogeneous (i.e., with differing κ_i and/or t_i^*), the weighted sum $\Delta(t)$ cannot be constant in t unless all $\alpha_i(t)$ are identical, which they are not under the model's assumptions. Therefore, $\Delta(t) \neq 0$. By continuity of $\Delta(t)$, it follows that

$$\exists t > 0 \text{ such that } \Delta(t) \neq 0 \quad \Rightarrow \quad \gamma_{j_1}(t) \neq \gamma_{j_2}(t). \quad (32)$$

Hence, automation exposure is strictly asymmetric across labor classes with differing sectoral exposure profiles. $\square$

Proposition 3.1 formalizes how differences in sectoral employment patterns (β_j) translate into heterogeneous exposure to automation ($\gamma_j(t)$) across labor classes. The exposure function $\gamma_j(t) = \sum_i \beta_{ij} \alpha_i(t)$ is a convex combination of sectoral automation intensities $\alpha_i(t)$, with weights β_{ij} . These weights reflect how much of labor class j 's employment is concentrated in sector i . Since the AGI intensity $\alpha_i(t)$ evolves differently across sectors due to differing adoption rates (κ_i) and timing (t_i^*), classes with different sectoral exposure weights will experience different composite automation trajectories. For instance, consider two classes j_1 and j_2 , with $\beta_{i^*j_1} > \beta_{i^*j_2}$ for a sector i^* that automates early (small $t_{i^*}^*$) or rapidly (large κ_{i^*}). Then, the exposure path $\gamma_{j_1}(t)$ will rise faster than $\gamma_{j_2}(t)$, implying earlier and more intense displacement pressure on class j_1 . Economically, this result captures the essence of differential automation risk: labor classes more concentrated in early-adopting or fast-automating sectors face quicker erosion of wage income. As $\gamma_j(t)$ enters directly into each household's

budget constraint (cf. Equation (16)), it reduces the effective labor income $(1 - \gamma_j(t))w_j(t)$, tightening consumption possibilities unless offset by redistribution. This creates urgency for targeted fiscal responses like class-calibrated UBI. The proposition thus provides the theoretical foundation for class-specific displacement dynamics and justifies asymmetric policy design in response to technological diffusion.

Proposition 3.2 (VAT Capture of AGI-Attributed Output Value). *Let sector i produce output $Q_i(t)$ via the nested CES production function defined in equation (8), where AGI labor enters as*

$$L_{i,AGI}(t) = \psi_i \alpha_i(t) K_i^{(AGI)}(t), \quad (33)$$

with $\psi_i > 0$ representing AGI productivity, $\alpha_i(t) \in (0, 1)$ the automation intensity, and $K_i^{(AGI)}(t)$ the AGI capital stock.

Assume a VAT is levied at rate $\theta_i \in [0, 1]$ on the revenue share attributed to AGI labor in sector i . If the marginal product of AGI labor is

$$MP_i^{AGI}(t) := \frac{\partial Q_i(t)}{\partial L_{i,AGI}(t)}, \quad (34)$$

then total VAT revenue from AGI-enabled production in sector i is

$$VAT_i(t) = \theta_i \cdot P_i(t) \cdot MP_i^{AGI}(t) \cdot L_{i,AGI}(t). \quad (35)$$

Setting $\theta_i = 1$ implies full taxation of AGI labor's marginal revenue product.

Proof. From equation (33), AGI labor in sector i is given by

$$L_{i,AGI}(t) = \psi_i \alpha_i(t) K_i^{(AGI)}(t).$$

Assuming perfect competition and constant returns to scale, factor payments equal marginal products times quantities. The marginal product of AGI labor is

$$MP_i^{AGI}(t) = \frac{\partial Q_i(t)}{\partial L_{i,AGI}(t)},$$

and the corresponding income contribution of AGI labor is

$$\text{Rev}_i^{AGI}(t) := P_i(t) \cdot MP_i^{AGI}(t) \cdot L_{i,AGI}(t).$$

Imposing a VAT at rate θ_i on this AGI-attributed revenue yields:

$$VAT_i(t) = \theta_i \cdot \text{Rev}_i^{AGI}(t) = \theta_i \cdot P_i(t) \cdot MP_i^{AGI}(t) \cdot L_{i,AGI}(t),$$

as stated. When $\theta_i = 1$, the VAT captures the full marginal revenue product of AGI labor. $\square$

Proposition 3.2 establishes that a value-added tax (VAT) levied on the marginal revenue product of AGI labor provides a tractable mechanism for capturing the automation-induced surplus in production. By linking the tax base directly to AGI-attributed output, the formulation ensures that as AGI substitutes for human labor, the fiscal base for redistribution shifts endogenously from wage income to automation-derived value. Setting $\theta_i = 1$ corresponds to fully taxing the marginal revenue generated by AGI labor, thereby creating a revenue-neutral foundation for financing universal basic income (UBI) transfers calibrated to the scale of labor displacement. Importantly, because the tax targets output rather than capital directly, it preserves investment incentives while maintaining the state's capacity to redistribute gains from automation as the labor share in production declines.

Proposition 3.3 (Targeted UBI Sustainability Under Budget Balance). *Let $UBI(t)$ denote the per-capita transfer to displaced workers at time t , and let*

$$N_{\text{disp}}(t) = \sum_j N_j \gamma_j(t) \quad (36)$$

denote the total number of displaced individuals. Let aggregate VAT revenue collected from all sectors be

$$R^{\text{VAT}}(t) = \sum_{i=1}^n \text{VAT}_i(t). \quad (37)$$

Then, under a balanced-budget policy in which all VAT revenue is redistributed exclusively to displaced individuals, a targeted UBI is fiscally sustainable if and only if

$$UBI(t) = \frac{R^{\text{VAT}}(t)}{N_{\text{disp}}(t)}. \quad (38)$$

Proof. The total fiscal expenditure on UBI at time t is given by

$$E^{\text{UBI}}(t) = N_{\text{disp}}(t) \cdot UBI(t), \quad (39)$$

where $N_{\text{disp}}(t)$ is the number of displaced individuals as defined in (36). Total VAT revenue is defined as

$$R^{\text{VAT}}(t) = \sum_{i=1}^n \text{VAT}_i(t), \quad (40)$$

where each $\text{VAT}_i(t)$ denotes the revenue collected from AGI-enabled production in sector i . Under a pointwise balanced-budget condition—that is, assuming the government balances its budget at each point in time—we require

$$E^{\text{UBI}}(t) = R^{\text{VAT}}(t). \quad (41)$$

Substituting (39) into (41) yields

$$N_{\text{disp}}(t) \cdot UBI(t) = R^{\text{VAT}}(t).$$

Solving for $UBI(t)$, we obtain

$$UBI(t) = \frac{R^{\text{VAT}}(t)}{N_{\text{disp}}(t)}. \quad (42)$$

This identity must hold for UBI to be exactly funded by VAT revenue. Conversely, if (38) holds, then government expenditures exactly match revenues by construction. Therefore, the condition is both necessary and sufficient for targeted UBI sustainability under a balanced-budget policy. $\square$

Equation (38) formalizes the direct relationship between automation-induced fiscal surplus and income replacement for technologically displaced workers. Crucially, the formula indicates that per-capita UBI is a function not of the total population, but of the mass of displaced labor $N_{\text{disp}}(t)$ and the size of the VAT base. This formulation ensures that the economic gains from AGI deployment are recycled specifically to those bearing the costs of displacement, thereby operationalizing a principle of targeted fiscal fairness without diluting transfers across the broader population.

However, the structure also introduces an inherent trade-off: as displacement intensifies—reflected in higher $\gamma_j(t)$ across labor classes—the denominator in (38) increases, which may depress per-worker UBI unless VAT revenues grow proportionally. This dynamic creates potential volatility in redistribution outcomes and underscores the importance of adaptive policy design.

In this context, several fiscal mechanisms become critical. First, VAT rates may need to adjust endogenously over time, or the AGI-related tax base may need to be broadened to maintain revenue adequacy. Second, fiscal buffers or countercyclical borrowing can help smooth UBI payments across periods of rapid automation. Third, integrating automation monitoring into fiscal planning processes ensures that real-time labor market data informs tax-and-transfer calibration. Collectively, these measures support the sustainability and equity of the redistribution scheme as the economy transitions toward higher levels of automation.

Equation (38) offers a transparent and operationalizable mechanism for aligning the fiscal incidence of automation with its distributive consequences, thereby providing a foundation for automation-responsive social policy.

Proposition 3.4 (Declining Labor Income Share with AGI Substitution). *Consider a general equilibrium economy in which each sector i produces output $Q_i(t)$ according to the nested CES function in Equation (8), combining effective human labor $L_{i,h}(t)$ and AGI labor $L_{i,AGI}(t)$ with elasticity parameter $\rho_1 < 1$. Suppose:*

- (i) *AGI intensity satisfies $\alpha_i(t) \rightarrow 1$ as $t \rightarrow \infty$ for all i (Equation (3)),*
- (ii) *Human labor supply declines such that $L_{i,h}(t) \rightarrow 0$ as $t \rightarrow \infty$ (Equations (5), (6)),*
- (iii) *Wages are determined by the marginal product of human labor (Equation (18)),*
- (iv) *Aggregate output $\sum_i Q_i(t)$ remains strictly positive as $t \rightarrow \infty$.*

Then, the aggregate labor income share converges to zero

$$\lim_{t \rightarrow \infty} \frac{\sum_i \frac{\partial Q_i(t)}{\partial L_{i,h}(t)} \cdot L_{i,h}(t)}{\sum_i Q_i(t)} = 0. \quad (43)$$

Proof. From Equation (5), effective human labor in sector i is given by

$$L_{ij}(t) = \beta_{ij} N_j (1 - \gamma_j(t)).$$

As AGI intensity $\alpha_i(t) \rightarrow 1$, automation exposure $\gamma_j(t) \rightarrow 1$, implying

$$L_{i,h}(t) = \sum_j L_{ij}(t) \rightarrow 0 \quad (\text{by Equation (6)})$$

AGI labor evolves as

$$L_{i,AGI}(t) = \psi_i \alpha_i(t) K_i^{(AGI)}(t),$$

and since $\alpha_i(t) \rightarrow 1$ and $K_i^{(AGI)}(t)$ grows over time (per Equation (13)), the AGI labor input becomes large. In the CES aggregator with $\rho_1 < 1$, human and AGI labor are gross substitutes. As $L_{i,h}(t) \rightarrow 0$ and $L_{i,AGI}(t)$ increases, the marginal product of human labor $\frac{\partial Q_i(t)}{\partial L_{i,h}(t)} \rightarrow 0$. Therefore, sectoral labor income satisfies

$$\frac{\partial Q_i(t)}{\partial L_{i,h}(t)} \cdot L_{i,h}(t) \rightarrow 0.$$

Summing across all sectors,

$$\sum_i \frac{\partial Q_i(t)}{\partial L_{i,h}(t)} \cdot L_{i,h}(t) \rightarrow 0.$$

By assumption, total output $\sum_i Q_i(t)$ remains bounded away from zero. Hence, the aggregate labor income share converges to zero

$$\lim_{t \rightarrow \infty} \frac{\sum_i \frac{\partial Q_i(t)}{\partial L_{i,h}(t)} \cdot L_{i,h}(t)}{\sum_i Q_i(t)} = 0. \quad \square$$

□

Proposition 3.4 formalizes a central implication of AGI-driven automation: as human labor becomes increasingly displaced (cf. Equation (25)), the share of economic output accruing to workers asymptotically vanishes. The nested CES production structure in Equation (8) implies that even when human and AGI labor are imperfect substitutes ($\rho_1 < 1$), the effective abundance and rising productivity of AGI labor enable it to substitute for human input at scale. As AGI intensity rises and human labor supply contracts, the marginal product of human labor declines (Equation (18)), driving wages toward zero despite continued or even increasing aggregate output. In the long-run equilibrium, value-added becomes almost entirely attributable to AGI-enabled capital services.

In the model framework, households do not directly own physical or AGI-specific capital (see discussion following Equation (17)), implying that absent redistribution, displaced workers face a collapse in income and consumption. This result underscores both the fiscal and normative rationale for automation-contingent transfers, such as the VAT-financed, targeted universal basic income (UBI) mechanism introduced in Equations (19)–(22). As the labor income share declines, a robust and responsive social safety net becomes essential to sustain household welfare and economic stability in the face of structural transformation.

Proposition 3.5 (S-Curve Dynamics of Labor Displacement). *Let AGI intensity in sector i evolve according to the logistic diffusion function (Equation (3))*

$$\alpha_i(t) = \frac{1}{1 + e^{-\kappa_i(t-t_i^*)}}, \quad \kappa_i > 0. \quad (44)$$

Define the automation exposure of labor class j as

$$\gamma_j(t) = \sum_{i=1}^n \beta_{ij} \alpha_i(t), \quad \text{with } \sum_i \beta_{ij} = 1, \quad (45)$$

and let the effective labor supplied by class j to sector i be

$$L_{ij}(t) = \beta_{ij} N_j (1 - \gamma_j(t)). \quad (46)$$

Then the function $L_{ij}(t)$ exhibits an S-shaped trajectory over time, with curvature

$$\frac{d^2 L_{ij}(t)}{dt^2} \begin{cases} > 0 & \text{for } t < t^*, \\ = 0 & \text{for } t = t^*, \\ < 0 & \text{for } t > t^*, \end{cases} \quad (47)$$

where $t^* = \sum_i \beta_{ij} t_i^*$ is the weighted average of sectoral inflection points.

Proof. Differentiate $L_{ij}(t) = \beta_{ij} N_j (1 - \gamma_j(t))$ with respect to t :

$$\frac{dL_{ij}(t)}{dt} = -\beta_{ij} N_j \frac{d\gamma_j(t)}{dt}. \quad (48)$$

From the definition of $\gamma_j(t)$, we have

$$\frac{d\gamma_j(t)}{dt} = \sum_{k=1}^n \beta_{kj} \frac{d\alpha_k(t)}{dt}, \quad (49)$$

and using the derivative of the logistic function

$$\frac{d\alpha_k(t)}{dt} = \kappa_k \alpha_k(t) (1 - \alpha_k(t)). \quad (50)$$

Thus,

$$\frac{dL_{ij}(t)}{dt} = -\beta_{ij}N_j \sum_{k=1}^n \beta_{kj}\kappa_k\alpha_k(t)(1 - \alpha_k(t)). \quad (51)$$

Differentiating again

$$\frac{d^2L_{ij}(t)}{dt^2} = -\beta_{ij}N_j \sum_{k=1}^n \beta_{kj} \frac{d}{dt} [\kappa_k\alpha_k(t)(1 - \alpha_k(t))] \quad (52)$$

$$= -\beta_{ij}N_j \sum_{k=1}^n \beta_{kj}\kappa_k^2\alpha_k(t)(1 - \alpha_k(t))(1 - 2\alpha_k(t)). \quad (53)$$

Note that each term in the sum changes sign at $\alpha_k(t) = 0.5$, i.e., $t = t_k^*$, because

$$1 - 2\alpha_k(t) \begin{cases} > 0 & \text{for } t < t_k^*, \\ = 0 & \text{for } t = t_k^*, \\ < 0 & \text{for } t > t_k^*. \end{cases}$$

The overall sign of $\frac{d^2L_{ij}(t)}{dt^2}$ therefore switches from positive to negative as the weighted average of inflection points is crossed. Since $\gamma_j(t)$ is a convex combination of sigmoid functions $\alpha_k(t)$, its curvature changes at

$$t^* := \sum_k \beta_{kj}t_k^*,$$

yielding a global S-shaped transition for $L_{ij}(t)$, with inflection at t^* , as claimed. $\square$

Proposition 3.5 establishes that AGI-driven labor displacement follows a smooth, nonlinear trajectory characterized by S-curve dynamics. This adjustment path accelerates around a critical threshold t^* , defined as the weighted average of sector-specific automation inflection points. The S-shape arises from the logistic diffusion of AGI technologies within each sector (Equation (3)), which induces a convex-concave transition in automation exposure $\gamma_j(t)$, and hence in the effective labor supply $L_{ij}(t)$.

Economically, this implies that labor displacement unfolds unevenly over time. The process begins gradually, reflecting low initial AGI intensity, then accelerates sharply as automation technologies mature and diffuse, before tapering off as most displaceable human labor is substituted or marginalized. The turning point t^* thus marks the period of maximum displacement intensity.

From a policy perspective, this curvature has direct implications for the timing and design of automation-responsive redistribution. The inflection point t^* offers a forecast of when redistributive needs—such as those addressed through targeted UBI transfers (Equation (22))—will reach their peak. Anticipating the nonlinear dynamics of $L_{ij}(t)$ is therefore essential for constructing fiscally adequate and temporally responsive interventions to mitigate sudden declines in labor income and to support macroeconomic stability during periods of accelerated automation.

Proposition 3.6 (Rising Income Inequality Under AGI Capital Concentration). *Let total income in the economy be divided between labor income and AGI-based capital income. Assume that:*

Labor income accrues to the broad working population and is given by

$$Y_L(t) = \sum_i w_i(t) L_{i,h}(t), \quad (54)$$

where $w_i(t)$ denotes the wage in sector i , and $L_{i,h}(t)$ is the effective supply of human labor (as in Equations (18) and (6)).

AGI capital income accrues to a fixed, small share of agents and is given by

$$Y_K(t) = \sum_i r_{AGI}(t) K_i^{(AGI)}(t), \quad (55)$$

where $K_i^{(AGI)}(t)$ evolves according to Equation (13).

Then, in the absence of redistribution and under AGI diffusion $\alpha_i(t) \rightarrow 1$, the Gini coefficient $G(t)$ rises over time:

$$\frac{dG(t)}{dt} > 0 \quad \text{as } t \rightarrow \infty. \quad (56)$$

Proof. Total income in the economy at time t is:

$$Y(t) = Y_L(t) + Y_K(t). \quad (57)$$

From Equation (6), human labor in sector i is:

$$L_{i,h}(t) = \sum_j \beta_{ij} N_j (1 - \gamma_j(t)), \quad (58)$$

where $\gamma_j(t) = \sum_i \beta_{ij} \alpha_i(t)$ increases as AGI intensity $\alpha_i(t) \rightarrow 1$ (see Equation (3)). Therefore:

$$\frac{d\gamma_j(t)}{dt} > 0 \quad \Rightarrow \quad \frac{dL_{i,h}(t)}{dt} < 0. \quad (59)$$

Assuming wages $w_i(t)$ are non-increasing or weakly elastic to labor input, labor income declines:

$$\frac{dY_L(t)}{dt} = \sum_i \left[\frac{dw_i(t)}{dt} L_{i,h}(t) + w_i(t) \frac{dL_{i,h}(t)}{dt} \right] < 0. \quad (60)$$

Simultaneously, AGI capital income increases with rising capital stock:

$$\frac{dY_K(t)}{dt} > 0, \quad (61)$$

given sustained growth in $K_i^{(AGI)}(t)$ and persistently positive returns $r_{AGI}(t)$.

Since capital ownership is concentrated while labor income is broadly distributed, the income distribution becomes increasingly unequal. The Gini coefficient $G(t)$ therefore rises over time, as formalized in Equation (56). $\square$

Equation (56) formalizes a structural dynamic: labor income $Y_L(t)$ declines (per Equation (60)) while capital income $Y_K(t)$ increases (see Equation (61)), leading to growing income concentration. As AGI labor substitutes for human labor and firms accumulate AGI capital (via Equation (13)), income increasingly accrues to capital holders. In the absence of redistribution—such as targeted UBI and AGI-linked VAT as defined in Equations (22) and (20)—this transition leads to a mechanically increasing Gini coefficient. The result underscores the need for automation-contingent fiscal tools to preserve distributional equity in post-labor economies.

Proposition 3.7 (Targeted UBI Fully Offsets AGI-Induced Labor Income Loss). *Let $w_j(t)$ denote the wage of labor class j , as defined in Equation (18), and let $\gamma_j(t) \in [0, 1]$ be the automation exposure of class j , from Equation (25). Define per-capita labor income for class j as*

$$y_j(t) = (1 - \gamma_j(t))w_j(t). \quad (62)$$

Suppose the government collects total VAT revenue,

$$R^{\text{VAT}}(t) = \sum_i T_i(t), \quad (63)$$

as defined in Equation (20), and distributes it equally among the displaced population via a targeted UBI

$$\text{UBI}(t) = \begin{cases} \frac{R^{\text{VAT}}(t)}{N_{\text{disp}}(t)}, & \text{if } N_{\text{disp}}(t) > 0, \\ 0, & \text{otherwise,} \end{cases} \quad (64)$$

where $N_{\text{disp}}(t) = \sum_j N_j \gamma_j(t)$ is the total displaced population (Equation (21)).

Then, a sufficient condition for UBI to fully offset AGI-induced labor income losses for all displaced individuals is

$$\text{UBI}(t) \geq \max_j [\gamma_j(t)w_j(t)]. \quad (65)$$

Under this condition, the total income of every class j satisfies

$$y_j(t) + \text{UBI}(t) \geq w_j(t), \quad (66)$$

i.e., no group experiences a reduction in income relative to their pre-automation wage baseline.

Proof. Per Equation (62), wage income for class j at time t is given by

$$y_j(t) = (1 - \gamma_j(t))w_j(t).$$

The corresponding income loss due to automation is

$$\Delta_j(t) = w_j(t) - y_j(t) = \gamma_j(t)w_j(t).$$

Suppose UBI satisfies the sufficient condition in Equation (65)

$$\text{UBI}(t) \geq \max_j \Delta_j(t).$$

Then, for every class j ,

$$\text{UBI}(t) \geq \Delta_j(t).$$

Adding UBI to the remaining labor income yields

$$y_j(t) + \text{UBI}(t) \geq y_j(t) + \Delta_j(t) = w_j(t),$$

which confirms that total income does not fall below the pre-displacement level, proving Equation (66). $\square$

Proposition 3.7 demonstrates that a targeted universal basic income (UBI) can fully offset the loss in labor income caused by AGI-driven automation, provided the per-capita transfer is at least as large as the maximum wage loss across labor classes. This condition defines a sufficient upper bound for preserving individual income levels: if met, every worker's total income (wages plus UBI) remains no lower than it would have been absent automation.

Economically, this result offers a practical benchmark for designing redistribution policies aimed at neutralizing the distributive effects of technological displacement. However, its applicability is subject to two key limitations. First, the condition may not be fiscally feasible in scenarios of widespread displacement. If automation exposure $\gamma_j(t)$ is high across most classes, or if AGI deployment does not generate sufficiently large taxable surpluses, then the VAT revenue required to fund the prescribed UBI could exceed available resources. Second, the condition ensures income replacement but not welfare equivalence. Many workers may value employment for reasons beyond income—such as social identity, structure, or purpose—and the loss of work may impose utility costs that income alone cannot offset. Addressing these nonpecuniary losses would require complementary measures beyond monetary compensation.

Despite these caveats, Proposition 3.7 provides an analytically grounded threshold for automation-contingent redistribution. It underscores how the sustainability of fiscal responses to technological change hinges on the evolving interplay between labor displacement, AGI productivity, and the capacity of tax instruments to capture the surplus generated by automation.

Proposition 3.8 (AGI-VAT Revenue as a Substitute for Displaced Payroll Base). *Let $w_j(t)$ denote the wage of labor class j at time t , and $\gamma_j(t)$ denote the automation exposure of class j , as defined in Equation (25). Then, the total displaced labor income at time t is*

$$W^{\text{loss}}(t) = \sum_j N_j \gamma_j(t) w_j(t), \quad (67)$$

where N_j is the size of class j . Let AGI labor input in sector i be defined via Equation (7)

$$L_{i,\text{AGI}}(t) = \psi_i \alpha_i(t) K_i^{(\text{AGI})}(t), \quad (68)$$

and define the marginal product of AGI labor as

$$MP_i^{\text{AGI}}(t) = \frac{\partial Q_i(t)}{\partial L_{i,\text{AGI}}(t)}, \quad (69)$$

where $Q_i(t)$ is sectoral output from Equation (8).

Then the AGI-attributed output in sector i is

$$Q_i^{\text{AGI}}(t) = MP_i^{\text{AGI}}(t) \cdot L_{i,\text{AGI}}(t). \quad (70)$$

If the government levies a value-added tax (VAT) on AGI output at rate $\theta_i \in [0, 1]$ in each sector, then VAT revenue from sector i is

$$T_i(t) = \theta_i \cdot Q_i^{\text{AGI}}(t). \quad (71)$$

Total VAT revenue from AGI at time t is:

$$T(t) = \sum_i T_i(t). \quad (72)$$

Then a sufficient condition for the targeted UBI to fully offset displaced labor income is

$$T(t) \geq W^{\text{loss}}(t), \quad (73)$$

i.e., the VAT revenue from AGI labor equals or exceeds the total displaced wage bill.

Proof. The displaced wage income is given by Equation (67), which aggregates lost income from each labor class weighted by its exposure $\gamma_j(t)$. The AGI labor in sector i , given by Equation (68), contributes to output with marginal product $MP_i^{\text{AGI}}(t)$ per Equation (69), yielding the AGI-attributed component of production in Equation (70)

$$Q_i^{\text{AGI}}(t) = MP_i^{\text{AGI}}(t) \cdot L_{i,\text{AGI}}(t).$$

Applying a VAT at rate θ_i to this quantity gives sectoral revenue in Equation (71), and aggregating across sectors produces the total VAT revenue in Equation (72). If this total revenue satisfies the condition in Equation (73), then

$$T(t) = \sum_i \theta_i MP_i^{\text{AGI}}(t) \cdot L_{i,\text{AGI}}(t) \geq \sum_j N_j \gamma_j(t) w_j(t) = W^{\text{loss}}(t),$$

which implies that the AGI-VAT system generates enough fiscal capacity to fully compensate for displaced labor income. $\square$

Proposition 3.8 shows that under appropriate tax design, the government can fully reclaim the labor value lost to automation by taxing AGI's marginal contribution to production. This acts as a fiscal mirror of the declining payroll base. This condition underpins Equation (23), which guarantees that the targeted UBI is adequate to replace average foregone labor income. Unlike payroll taxes, which shrink as labor is displaced, AGI-linked VAT dynamically scales with automation intensity (via $\alpha_i(t)$) and capital deepening ($K_i^{(\text{AGI})}(t)$), maintaining fiscal adequacy in the face of structural change.

Proposition 3.9 (Existence of a Post-Labor Balanced Growth Path). *Consider an economy where human labor is fully displaced, i.e., $L_{i,h}(t) = 0$ for all sectors i and times t , and AGI labor is fully active such that $\alpha_i(t) \rightarrow 1$ as $t \rightarrow \infty$, per Equation (3). Let AGI labor be defined by*

$$L_{i,AGI}(t) = \psi_i \alpha_i(t) K_i^{(AGI)}(t). \quad (74)$$

Assume sectoral output is given by a CES function over AGI capital and AGI-derived labor

$$Q_i(t) = A_i \left[a_i \left(K_i^{(AGI)}(t) \right)^\rho + (1 - a_i) (L_{i,AGI}(t))^\rho \right]^{\frac{1}{\rho}}, \quad (75)$$

with $a_i \in (0, 1)$, $\rho < 1$, and total factor productivity $A_i > 0$. Let capital accumulate according to

$$\dot{K}_i^{(AGI)}(t) = s_i Q_i(t) - \delta_i K_i^{(AGI)}(t), \quad (76)$$

where $s_i \in (0, 1)$ is the savings rate and $\delta_i > 0$ is the depreciation rate.

Then there exists a constant growth rate $g_i > 0$ such that, along the balanced growth path (BGP),

$$\frac{\dot{Q}_i(t)}{Q_i(t)} = \frac{\dot{K}_i^{(AGI)}(t)}{K_i^{(AGI)}(t)} = g_i, \quad (77)$$

i.e., both capital and output grow at the same exponential rate in the post-labor economy.

Proof. As $\alpha_i(t) \rightarrow 1$, Equation (74) implies

$$L_{i,AGI}(t) \rightarrow \psi_i K_i^{(AGI)}(t). \quad (78)$$

Substitute this into Equation (75)

$$\begin{aligned} Q_i(t) &= A_i \left[a_i \left(K_i^{(AGI)}(t) \right)^\rho + (1 - a_i) \left(\psi_i K_i^{(AGI)}(t) \right)^\rho \right]^{\frac{1}{\rho}} \\ &= A_i [a_i + (1 - a_i) \psi_i^\rho]^{\frac{1}{\rho}} K_i^{(AGI)}(t). \end{aligned} \quad (79)$$

Define the constant

$$\Xi_i := A_i [a_i + (1 - a_i) \psi_i^\rho]^{\frac{1}{\rho}} > 0, \quad (80)$$

so that Equation (79) becomes

$$Q_i(t) = \Xi_i K_i^{(AGI)}(t). \quad (81)$$

Substitute into the capital accumulation equation (76):

$$\begin{aligned} \dot{K}_i^{(AGI)}(t) &= s_i \Xi_i K_i^{(AGI)}(t) - \delta_i K_i^{(AGI)}(t) \\ &= (s_i \Xi_i - \delta_i) K_i^{(AGI)}(t). \end{aligned} \quad (82)$$

Define the net growth rate

$$g_i := s_i \Xi_i - \delta_i. \quad (83)$$

If $g_i > 0$, then $K_i^{(AGI)}(t) = K_{i0} e^{g_i t}$ and, from Equation (81), $Q_i(t) = \Xi_i K_{i0} e^{g_i t}$. Hence, Equation (77) follows. $\square$

Proposition 3.9 establishes that even in the complete absence of human labor, the economy can sustain a balanced growth path. AGI labor services, as defined in Equation (74), become fully capital-embodied, and the CES production function reduces to a linear form in AGI capital (Equation (81)). The endogenous growth rate g_i , derived in Equation (83), depends solely on productivity-adjusted savings and depreciation, decoupling growth from demographic or labor-market constraints. This result validates the internal consistency of the model's post-labor regime: the economy does not collapse with labor displacement, but instead converges to a purely capital-driven path of exponential growth.

Proposition 3.10 (Transitional Disequilibrium Due to AGI Capital Adjustment Costs).

Consider an economy where AGI-specific capital $K_i^{(AGI)}(t)$ accumulates via investment $I_i^{(AGI)}(t)$, subject to convex adjustment costs given by

$$\phi_i \left(I_i^{(AGI)}(t), K_i^{(AGI)}(t) \right) = \frac{\vartheta}{2} \left(\frac{I_i^{(AGI)}(t)}{K_i^{(AGI)}(t)} \right)^2 K_i^{(AGI)}(t), \quad (84)$$

as defined in Equation (14). Let the aggregate resource constraint be

$$\sum_i Q_i(t) = \sum_j N_j C_j(t) + \sum_i I_i^{(AGI)}(t) + \sum_i \phi_i \left(I_i^{(AGI)}(t), K_i^{(AGI)}(t) \right), \quad (85)$$

as in Equation (??), and assume output $Q_i(t)$ is increasing in automation intensity $\alpha_i(t)$ (see Equation (3)).

Then, during the transition to AGI-intensive production, there exists a time interval over which

$$\frac{d}{dt} \left(\sum_j N_j C_j(t) \right) < 0, \quad (86)$$

i.e., aggregate consumption declines temporarily—even as output rises—due to reallocation of resources toward costly AGI capital adjustment.

Proof. From the resource constraint (85), define aggregate consumption

$$C_{\text{agg}}(t) := \sum_j N_j C_j(t) = \sum_i Q_i(t) - \sum_i \left[I_i^{(AGI)}(t) + \phi_i \left(I_i^{(AGI)}(t), K_i^{(AGI)}(t) \right) \right]. \quad (87)$$

Differentiating both sides with respect to time

$$\frac{dC_{\text{agg}}(t)}{dt} = \sum_i \frac{dQ_i(t)}{dt} - \sum_i \left[\frac{dI_i^{(AGI)}(t)}{dt} + \frac{d}{dt} \phi_i \left(I_i^{(AGI)}(t), K_i^{(AGI)}(t) \right) \right]. \quad (88)$$

By the chain rule, the time derivative of the adjustment cost function is

$$\frac{d}{dt} \phi_i = \frac{\partial \phi_i}{\partial I_i} \cdot \dot{I}_i^{(AGI)}(t) + \frac{\partial \phi_i}{\partial K_i} \cdot \dot{K}_i^{(AGI)}(t), \quad (89)$$

where

$$\frac{\partial \phi_i}{\partial I_i} = \vartheta \cdot \frac{I_i^{(\text{AGI})}(t)}{K_i^{(\text{AGI})}(t)}, \quad (90)$$

$$\frac{\partial \phi_i}{\partial K_i} = \frac{\vartheta}{2} \left[\left(\frac{I_i^{(\text{AGI})}(t)}{K_i^{(\text{AGI})}(t)} \right)^2 - 2 \cdot \left(\frac{I_i^{(\text{AGI})}(t)}{K_i^{(\text{AGI})}(t)} \right)^2 \right] = -\frac{\vartheta}{2} \left(\frac{I_i^{(\text{AGI})}(t)}{K_i^{(\text{AGI})}(t)} \right)^2. \quad (91)$$

If investment grows rapidly during automation (e.g., due to falling AGI prices or anticipated productivity gains), then $\dot{I}_i^{(\text{AGI})}(t)$ and $\dot{K}_i^{(\text{AGI})}(t)$ are large, causing the adjustment cost derivative in (89) to dominate (88). Therefore, even when $\frac{dQ_i(t)}{dt} > 0$, we may still have

$$\frac{dC_{\text{agg}}(t)}{dt} < 0, \quad (92)$$

as claimed in Equation (86). $\square$

Proposition 3.10 illustrates that technological progress via AGI may entail a short-run decline in household consumption. This results from nonlinear capital adjustment costs (84) that absorb a growing share of output during investment surges. Although sectoral output $Q_i(t)$ rises due to increasing AGI intensity (cf. Equation (7)), the economy may experience a temporary welfare dip as consumption is crowded out by costly automation scaling. This transitional friction reinforces the role of redistributive buffers—such as targeted UBI (Equation (22))—in maintaining household resilience during AGI-induced structural transformations.

Proposition 3.11 (Welfare Gains from AGI-Funded Targeted UBI). *Let household j 's lifetime utility be given by the standard discounted logarithmic form*

$$U_j = \int_0^\infty e^{-\rho t} \log(C_j(t)) dt, \quad (93)$$

as defined in Equation (15), where $\rho > 0$ is the subjective discount rate and $C_j(t)$ denotes consumption at time t .

Let $C_j^{\text{UBI}}(t)$ and $C_j^{\text{LF}}(t)$ denote the consumption paths faced by household j under two regimes. In the targeted UBI regime, displaced workers receive transfers from AGI-linked VAT revenue, as specified in Equation (22). In the laissez-faire regime, there is no redistribution, and $\text{UBI}(t) = 0$.

If

$$C_j^{\text{UBI}}(t) \geq C_j^{\text{LF}}(t) \quad \text{for all } t \geq 0, \quad (94)$$

then lifetime utility is weakly higher under UBI

$$U_j^{\text{UBI}} \geq U_j^{\text{LF}}. \quad (95)$$

Furthermore, if there exists a measurable set $S \subset [0, \infty)$ with positive measure such that

$$C_j^{\text{UBI}}(t) > C_j^{\text{LF}}(t) \quad \text{for all } t \in S, \quad (96)$$

then lifetime utility is strictly higher

$$U_j^{\text{UBI}} > U_j^{\text{LF}}. \quad (97)$$

Proof. From Equation (93), utility is strictly increasing and strictly concave in consumption. Under the assumption that $C_j^{\text{UBI}}(t) \geq C_j^{\text{LF}}(t)$ for all t , it follows from the monotonicity of the logarithmic function that $\log(C_j^{\text{UBI}}(t)) \geq \log(C_j^{\text{LF}}(t))$ for all $t \geq 0$. Multiplying by the positive discount factor $e^{-\rho t}$ and integrating over time yields inequality (95).

If, in addition, $C_j^{\text{UBI}}(t) > C_j^{\text{LF}}(t)$ on a set $S \subset [0, \infty)$ with $\mu(S) > 0$, then the integrand $e^{-\rho t} \log(C_j^{\text{UBI}}(t))$ strictly dominates $e^{-\rho t} \log(C_j^{\text{LF}}(t))$ over that set. The inequality in Equation (97) follows by integrating over a strictly positive integrand on a set of nonzero measure. $\square$

Proposition 3.11 establishes a sufficient condition under which a targeted UBI policy funded by AGI-linked VAT increases intertemporal welfare for displaced workers. Since automation exposure $\gamma_j(t)$ reduces wage income and, consequently, consumption (cf. Equation (16)), transfers that offset this loss help smooth consumption trajectories over time.

Under logarithmic utility, even modest increases in consumption—particularly during early periods when discounting has less effect—generate significant gains in lifetime utility. Thus, this result highlights that automation-contingent redistribution is not only equity-enhancing but also welfare-improving in a dynamic sense. In particular, UBI serves not merely as compensation but as an effective mechanism for sustaining household utility during structural labor displacement.

4 Discussion and Policy Implications

The analytical results developed in this paper contribute to a growing body of research on the macroeconomic consequences of automation and the fiscal architecture needed to manage its distributional impact. By formalizing the dynamics of labor displacement, income concentration, and redistribution under AGI diffusion, the model offers a unified framework for designing equitable and sustainable policy in an increasingly post-labor economy.

The asymmetric exposure of labor classes to automation, shown in Proposition 3.1, arises endogenously from heterogeneity in sectoral employment shares and substitutability with AGI labor. This generalizes the empirical observations of Autor (2015) and Acemoglu and Restrepo (2019), who document uneven displacement across skill levels and occupations. Proposition 3.5 further refines this view by showing that displacement follows a logistic time path, consistent with the S-curve adoption dynamics typical of general-purpose technologies (Comin and Hobijn, 2010). These findings imply that automation exposure is not only

unequal but temporally concentrated, with identifiable inflection points—highlighting the importance of timely, class-specific interventions rather than universal or static policies.

At the fiscal level, Propositions 3.2 and 3.8 identify a novel mechanism for capturing automation surplus through a value-added tax (VAT) on AGI-generated output. This approach parallels recent proposals by Korinek (2021) and Guner et al. (2023), who argue for automation-sensitive tax instruments that do not penalize labor or disincentivize innovation. By aligning taxation with AGI labor's marginal productivity, the VAT acts as a Pigouvian instrument: it internalizes the social cost of displacement while preserving efficiency at the margin. Crucially, this allows for revenue generation without distorting either capital formation or remaining labor markets—an increasingly rare property in the design of 21st-century tax systems.

Building on this fiscal foundation, Propositions 3.3 and 3.7 derive formal sufficiency conditions under which targeted UBI transfers can restore income lost to automation. This advances the normative arguments made by Bidadanure (2019) and Farhi and Werning (2020), translating them into concrete policy thresholds. Targeting UBI to displaced workers rather than providing a universal transfer enhances fiscal efficiency while preserving equity, avoiding the regressivity that often plagues untargeted UBI schemes. The model thus provides a dynamic, distribution-aware rationale for income maintenance that is both theoretically grounded and practically implementable.

Proposition 3.6 directly links the rise in capital income from AGI adoption to increasing income inequality, assuming capital is concentrated in a small subset of the population. This mirrors long-standing concerns in the macro-distribution literature—especially the capital-biased technological change hypothesis formalized by Acemoglu and Restrepo (2018) and the broader inequality dynamics explored by Piketty (2014). Proposition 3.11 complements this by demonstrating that redistributive transfers do not merely mitigate inequality but can lead to strict improvements in household utility, provided consumption gains are sustained over time. This is an important correction to the view that redistribution is purely compensatory; here, it functions as a dynamic welfare-enhancing instrument that stabilizes intertemporal consumption in the face of structural change.

The long-run viability of post-labor growth is affirmed in Proposition 3.9, which proves the existence of a balanced growth path even when human labor is fully displaced. Combined with Proposition 3.4, which shows the asymptotic vanishing of the labor share, the results suggest a fundamental macroeconomic transition: from a wage-based to a capital-based engine of growth. This contributes to the literature on endogenous growth under automation (e.g., Acemoglu and Restrepo (2022)) by showing that productivity growth can be sustained solely via AGI capital accumulation, decoupling economic output from human demographic trends.

However, this transition is not frictionless. Proposition 3.10 shows that convex AGI capital adjustment costs can temporarily depress household consumption even as output rises. This reflects a classic intertemporal misalignment—one with significant implications for social policy. As highlighted by Acemoglu and Restrepo (2022), transitions to new production paradigms often generate political resistance and welfare losses. The model shows that even in an efficiency-enhancing technological transition, welfare can fall temporarily unless buffered by redistributive mechanisms such as UBI. Here, redistribution plays a stabilizing role, not only correcting ex post inequality but also mitigating transitional consumption

shocks.

Finally, the existence of a dynamic general equilibrium, proven in Proposition A.1, affirms the internal coherence of the model. This permits its application in quantitative policy simulations, such as evaluating dynamic tax-transfer schemes or automation-triggered fiscal rules. The model's structure—combining sectoral CES production, endogenous displacement, convex capital adjustment, and targeted redistribution—offers a robust platform for calibrating AGI-era policy in both developed and emerging economies.

Taken together, the propositions offer a blueprint for macroeconomic governance in the face of rapid automation. The framework developed here links rigorous structural modeling to actionable fiscal tools, providing both a positive and normative account of how to manage the post-labor transition. It suggests that future policy should be guided by three key principles: responsiveness to technological diffusion, distributional targeting based on exposure, and fiscal instruments that scale with automation-generated surplus. By embedding these elements within general equilibrium theory, the model advances the theoretical and policy frontier on automation, inequality, and welfare.

5 Conclusion

This paper develops a dynamic general equilibrium model to analyze the macroeconomic and distributive consequences of artificial general intelligence (AGI) diffusion. Extending the task-based automation framework of Acemoglu and Restrepo (2018, 2019), the model introduces sectoral heterogeneity, nonlinear diffusion, capital adjustment frictions, and endogenous labor displacement. AGI capital substitutes for human labor asymmetrically across sectors and worker types, capturing the granular structure of modern automation transitions.

The central findings highlight that AGI adoption, while raising aggregate output, induces a structural decline in labor income and accelerates income concentration. Without policy intervention, this transition leads to persistently rising inequality—a result that echoes macro-distributional concerns in Piketty (2014) and Acemoglu and Restrepo (2022). To address this, the paper proposes a fiscal architecture centered on an AGI-linked value-added tax (VAT), whose revenue is redistributed as targeted universal basic income (UBI) to displaced workers.

This VAT-UBI scheme exhibits several attractive features. It is production-efficient, budget-balanced, and exposure-sensitive: it taxes value creation without disincentivizing innovation and redistributes benefits in proportion to automation-induced losses. These properties align with principles of optimal automation taxation in Korinek (2021) and Guner et al. (2023), and normative proposals for transition-era transfers in Farhi and Werning (2020) and Bidadanure (2019). The model also shows that targeted UBI can restore intertemporal household welfare and cushion transitional demand shocks, even when AGI diffusion is rapid and capital adjustment costly (cf. Hayashi, 1982).

Importantly, the framework abstracts from asset ownership and intergenerational dynamics to focus cleanly on fiscal redistribution. This isolates the role of tax-transfer systems but omits political economy and capital market mechanisms that shape inequality in real-world settings. Incorporating heterogeneous agents with savings, bequest motives, or voting power—as emphasized in Piketty (2014) and Korinek (2021)—is a key direction for future

research.

By unifying production-side automation dynamics with general equilibrium redistribution, the model offers a theoretically coherent and policy-relevant foundation for managing AGI-induced structural change. As economies decouple productive capacity from human labor, adaptive fiscal institutions become essential for preserving equity, stabilizing consumption, and maintaining social contract legitimacy.

Future work should explore how alternative institutional configurations—spanning taxation, ownership, and democratic governance—mediate the trajectory of AGI economies. In an era where labor may no longer be central to value creation, the design of redistribution systems will shape not only material outcomes but the political sustainability of growth itself.

A Existence of Equilibrium

Proposition A.1 (Existence of Dynamic General Equilibrium). *Consider the infinite-horizon economy defined in the main model, with heterogeneous sectors i and labor classes j , endogenous AGI displacement, and targeted fiscal redistribution. Under the assumptions specified—strictly concave household preferences, nested CES production technologies, convex AGI capital adjustment costs, smooth displacement and redistribution mappings, compactness of feasible allocations, and period-by-period government budget balance—there exists a dynamic general equilibrium in the sense of Definition 2.1. That is, there exists a sequence of time-indexed functions*

$$\{C_j(t), P_i(t), w_j(t), Q_i(t), K_i^{(H)}(t), K_i^{(AGI)}(t), T_i(t), \text{UBI}(t)\}_{i,j,t \geq 0},$$

satisfying household and firm optimization, market clearing, capital accumulation laws, and balanced fiscal redistribution.

Proof Sketch. We proceed in five steps, invoking standard results in general equilibrium theory (e.g., Lucas 1981, Mas-Colell et al. 1995, Stokey et al. 1989).

Step 1: Household optimization. Each household class maximizes lifetime utility subject to an intertemporal budget constraint, where labor income is reduced by automation exposure $\gamma_j(t)$, and UBI transfers depend on AGI-displacement. Preferences are strictly concave, feasible sets are convex and compact in appropriate function spaces, and solutions exist by the Weierstrass and Berge maximum theorems. The consumption path is continuous in prices and policy.

Step 2: Firm optimization. Each firm maximizes discounted profits subject to nested CES technologies and convex capital adjustment costs. The firm problem is concave in control variables, and production and adjustment mappings are smooth. The firm’s optimal policy is uniquely defined, and the input demand correspondence is continuous.

Step 3: Market clearing. Aggregated goods and factor markets clear. The excess demand correspondence is continuous, homogeneous of degree zero, satisfies Walras’ Law, and diverges at boundaries, satisfying conditions for equilibrium existence via Kakutani or Brouwer-type fixed point arguments.

Step 4: Government budget. The government collects VAT on AGI-attributed output and redistributes this revenue as targeted UBI. The transfer rule is continuous, bounded, and

satisfies a balanced budget identity pointwise. The fiscal mechanism is internally consistent with the production-displacement mapping.

Step 5: Fixed point. Define a mapping from prices and policies to allocations. The joint correspondence over prices, allocations, and taxes is upper hemicontinuous, convex-valued, and compact. Existence of a fixed point implies equilibrium. $\square$

B Cost Function Derivation via Duality

This appendix derives the firm's pricing rule in Equation (10) using duality theory, applied to the nested CES production structure specified in Equation (8). In particular, we derive the marginal cost $MC_i(t)$ via the cost minimization problem of a representative firm in sector i , conditional on producing a fixed level of output $Q_i(t)$ at given factor prices.

Firm's Cost Minimization Problem

Given an output target $Q_i(t)$, the representative firm solves

$$\begin{aligned} \min_{L_{i,h}(t), K_i^{(H)}(t), K_i^{(AGI)}(t)} \quad & C_i(t) = w_{i,h}(t)L_{i,h}(t) + r_H(t)K_i^{(H)}(t) + r_{AGI}(t)K_i^{(AGI)}(t) \\ \text{s.t.} \quad & Q_i(t) = A_i \left[\theta_1 \left(L_{i,h}(t)^{\rho_1} + \left(\psi_i \alpha_i(t) K_i^{(AGI)}(t) \right)^{\rho_1} \right)^{\frac{\eta_1}{\rho_1}} \right. \\ & \left. + \theta_2 \left(\left(K_i^{(H)}(t) \right)^{\rho_2} + \left(K_i^{(AGI)}(t) \right)^{\rho_2} \right)^{\frac{\eta_2}{\rho_2}} \right]^{\frac{1}{\eta}}. \end{aligned} \quad (98)$$

Here

- $w_{i,h}(t)$ is the effective wage in sector i , aggregating over labor classes using weights β_{ij} ;
- $r_H(t)$ and $r_{AGI}(t)$ are rental rates on traditional and AGI-specific capital, respectively;
- $\alpha_i(t)$ and ψ_i scale the AGI capital into effective labor services;
- A_i is sectoral total factor productivity;
- $\phi_i(\cdot)$ represents convex adjustment costs (excluded here as the problem is static in t).

Cost Function and Marginal Cost via Duality

Let the cost function be defined as the value function of the dual problem

$$C_i(Q_i(t), \mathbf{p}(t)) = \min_{\text{inputs}} \left\{ w_{i,h}(t)L_{i,h}(t) + r_H(t)K_i^{(H)}(t) + r_{AGI}(t)K_i^{(AGI)}(t) \mid Q_i(t) \text{ is produced} \right\}, \quad (99)$$

where $\mathbf{p}(t) = (w_{i,h}(t), r_H(t), r_{AGI}(t))$ is the vector of input prices. By standard results (e.g., Mas-Colell et al. 1995), the cost function is continuous, increasing, concave in output, and convex in input prices.

Shephard's Lemma. The marginal cost is the derivative of the cost function with respect to output

$$MC_i(t) = \frac{\partial C_i(Q_i(t), \mathbf{p}(t))}{\partial Q_i(t)}. \quad (100)$$

Pricing Rule. Under perfect competition, firms price at marginal cost:

$$P_i(t) = MC_i(t) = \frac{\partial C_i(Q_i(t), \mathbf{p}(t))}{\partial Q_i(t)}, \quad (101)$$

which corresponds to Equation (10) in the main text.

Interpretation and Application

The pricing condition in Equation (101) ensures that under constant returns to scale and competitive markets, firms earn zero economic profit. Although the nested CES form with AGI capital precludes a closed-form expression for $MC_i(t)$, numerical approximation is tractable using either envelope methods or perturbation techniques. The marginal cost depends crucially on

- The elasticity of substitution between human and AGI labor;
- The sectoral pace of AGI diffusion $\alpha_i(t)$ and its efficiency ψ_i ;
- The endogenous cost share of AGI capital, driven by its marginal productivity in labor-equivalent terms.

This dual approach formally connects micro-level production and factor pricing with macro-level price formation, thereby justifying the use of sectoral prices in the model's general equilibrium and fiscal modules.

Remark B.1 (On Computing $MC_i(t)$). Although the pricing rule is theoretically well-defined, practical implementation requires solving the cost minimization problem numerically at each time t . Due to the nested CES structure and AGI-related non-separabilities, closed-form expressions are intractable. Instead, standard methods such as interior-point optimization or finite-difference envelope approximation may be used. In steady-state analysis or transition path simulations, log-linearization around a balanced growth path provides accurate approximations for $MC_i(t)$. Thus, Equation (101) serves as a structurally grounded, computationally tractable pricing equation in automation-era general equilibrium models.

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